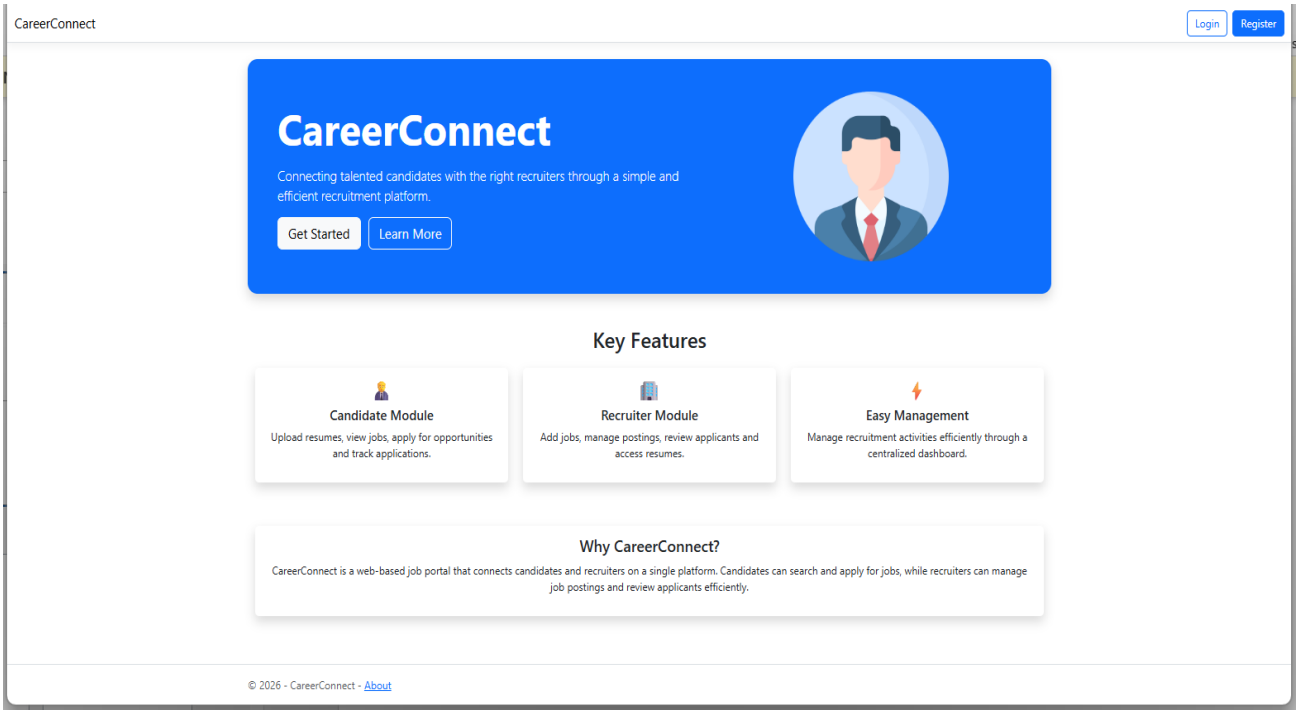
Project Name :-

CareerConnect-Job Portal Management System



From:- Aditya Sunil Garud

Manager:- Poonam Pawar

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ABSTRACT

CareerConnect is a web-based Job Portal Management System developed using ASP.NET Core MVC, Entity Framework Core, and MySQL. The system is designed to simplify the recruitment process by providing a common platform for both candidates and recruiters. In many organizations, recruitment activities such as job posting, resume collection, applicant tracking, and communication between recruiters and job seekers are often handled manually. These manual processes increase administrative effort and reduce overall recruitment efficiency.

The primary objective of CareerConnect is to digitize and streamline recruitment operations through a centralized web application. The platform provides separate modules for Candidates and Recruiters. Candidates can register, log in, upload resumes, browse available job opportunities, search jobs based on location, apply for suitable positions, and view details of previously applied jobs. Recruiters can register, log in, add new job postings, modify existing job information, delete obsolete vacancies, view applicants, and access uploaded resumes.

The application implements Session-Based Authentication and Role-Based Authorization to ensure secure access to system resources. Entity Framework Core is used for database management, while Bootstrap provides a responsive and user-friendly interface. The project demonstrates practical implementation of modern web development concepts including MVC Architecture, Authentication, Authorization, Database Management, File Upload Handling, and Recruitment Workflow Automation. CareerConnect serves as a complete job portal solution that improves recruitment efficiency and provides a seamless experience for both recruiters and job seekers.

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Software Development Life Cycle (SDLC)

Waterfall SDLC Model

The Software Development Life Cycle (SDLC) is a systematic process used for software development. It helps developers organize project activities into different phases and ensures that project requirements are successfully implemented.

For CareerConnect, the Waterfall Model was selected because the project requirements were clearly identified before development began. The model follows a sequential approach in which each phase is completed before moving to the next phase.

Waterfall Phases

- Planning
- Requirement Analysis
- System Design
- Development
- Testing
- Deployment
- Maintenance

Benefits of Using Waterfall Model for This Project

- Clear and well-defined development phases.
- Easy to understand and implement.
- Suitable for projects with fixed requirements.
- Provides proper documentation at each stage.
- Helps monitor project progress effectively.
- Reduces confusion during development.
- Ensures systematic implementation of project functionalities.

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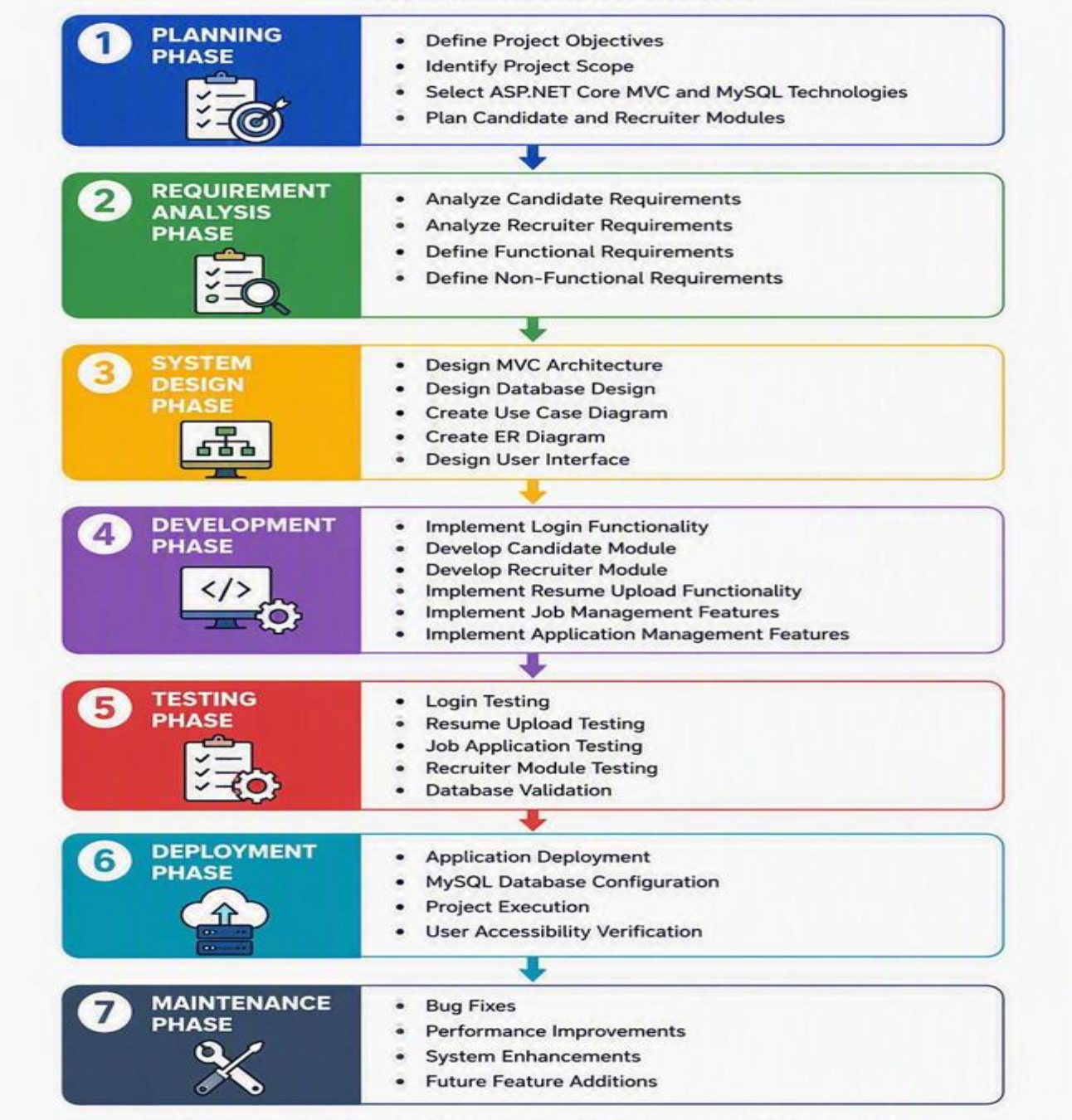


Figure 1: Waterfall SDLC Model Used for CareerConnect Job Portal Management

The Waterfall Model provided a structured approach for developing CareerConnect. Each phase was completed sequentially, which helped maintain project organization and ensured all requirements were properly implemented.

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Planning and Requirement Analysis

The planning phase focused on identifying recruitment-related challenges and defining the objectives of the system. The main goal was to provide a digital platform that allows candidates and recruiters to perform recruitment activities in an organized manner.

Traditional recruitment methods involve manual management of vacancies, resumes, and applications, making the hiring process time-consuming and difficult to track. To address these challenges, CareerConnect was planned as a centralized Job Portal Management System.

Candidate Requirements

- Register into the system
- Login securely
- Upload Resume
- View Available Jobs
- Search Jobs by Location
- Apply for Jobs
- View Applied Jobs
- View Job Details

Recruiter Requirements

- Register into the system
- Login securely
- Add Job Postings
- Manage Jobs
- Edit Job Information
- Delete Job Postings
- View Applicants
- Access Candidate Resumes

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Use Case Diagram

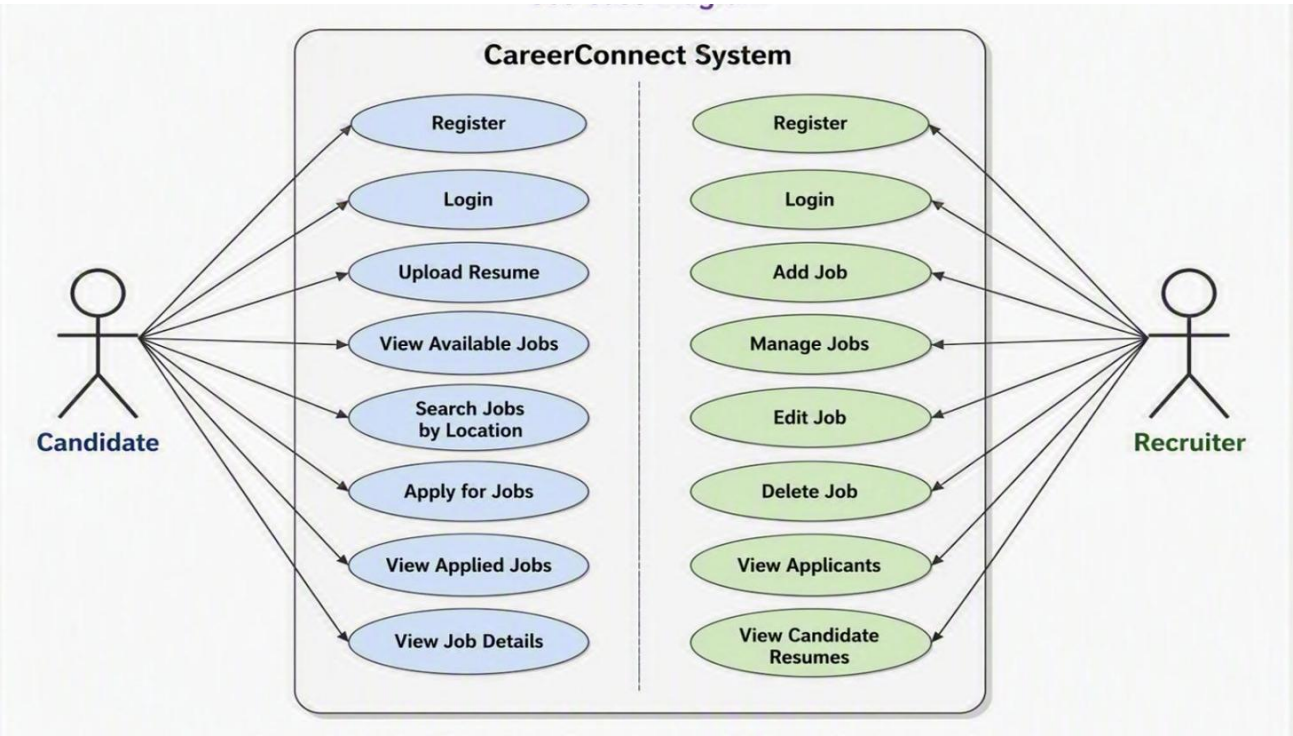


Figure 2: Use Case Diagram of CareerConnect Job Portal Management

The Use Case Diagram represents the interaction between the primary users of the CareerConnect system and the functionalities available to them. The system consists of two main actors, namely Candidate and Recruiter, each having specific roles and responsibilities within the recruitment process. Candidates can register, log in, upload resumes, search available jobs, apply for job opportunities, and view their application details, while Recruiters can create and manage job postings, review applicants, and access candidate resumes. The diagram provides a high-level overview of the system requirements and clearly illustrates how different users interact with the application to perform recruitment-related activities. It also helps in understanding the overall workflow and functionality of the CareerConnect Job Portal Management System.

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System Design

The System Design phase converts user requirements into a technical structure. CareerConnect was designed using the ASP.NET Core MVC architecture, which separates the application into Models, Views, and Controllers. This architecture improves maintainability, readability, and scalability of the application.

The View component is responsible for displaying data to users through a user-friendly interface. Controllers handle requests, process business logic, and communicate with the database through Models. Entity Framework Core is used as an ORM to manage database operations efficiently.

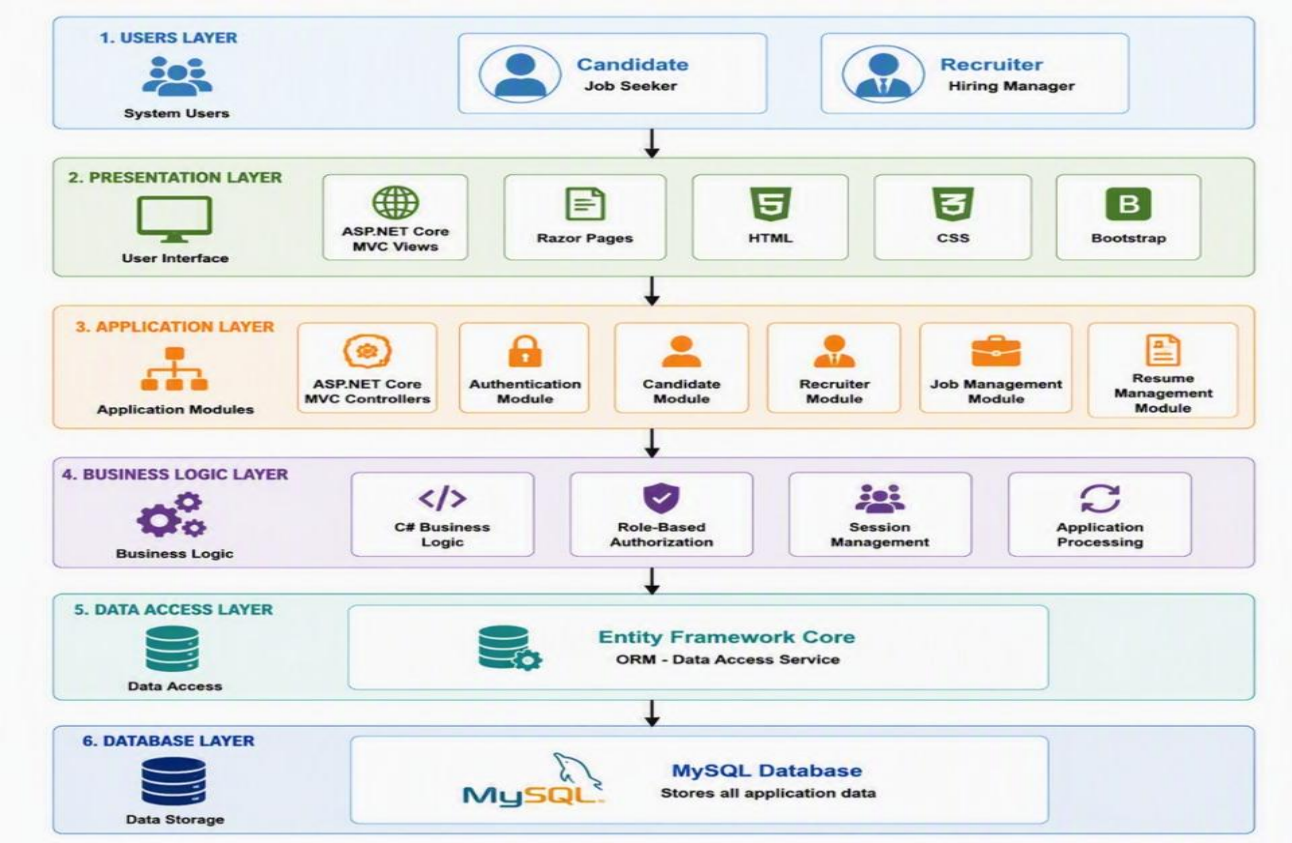


Figure 3: System Architecture Diagram of CareerConnect Job Portal Management

Technologies Used

The CareerConnect application was developed using ASP.NET Core MVC, Entity Framework Core, MySQL, HTML, CSS, Bootstrap, and Razor Views. HTML, CSS, and Bootstrap were used to design responsive and user-friendly interfaces, while Razor Views enabled dynamic content rendering within the MVC framework. ASP.NET Core MVC and C# were used to implement the application's core functionality and business logic. Entity Framework Core served as the data access layer, simplifying CRUD operations and database interactions. MySQL was used as the backend database for storing recruitment-related information such as users, jobs, resumes, and applications. These technologies collectively provided a robust foundation for building the CareerConnect Job Portal Management System.

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Database Design

The database design of CareerConnect was created to manage users, job postings, applications, and resume information effectively. A relational database structure was used to ensure data consistency and proper relationship management.

Main Tables

Users

Stores Candidate and Recruiter information.

Fields

- Id
- Name
- Email
- Password
- Role

Jobs

Stores job postings created by recruiters.

Fields

- Id
- Title
- Description
- RequiredSkills
- Experience
- Location
- RecruiterId

Applications

Stores records of candidate applications.

Fields

- Id
- CandidateId
- JobId
- AppliedDate

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Resumes

Stores uploaded resume information.

Fields

- Id
- CandidateId
- ResumePath

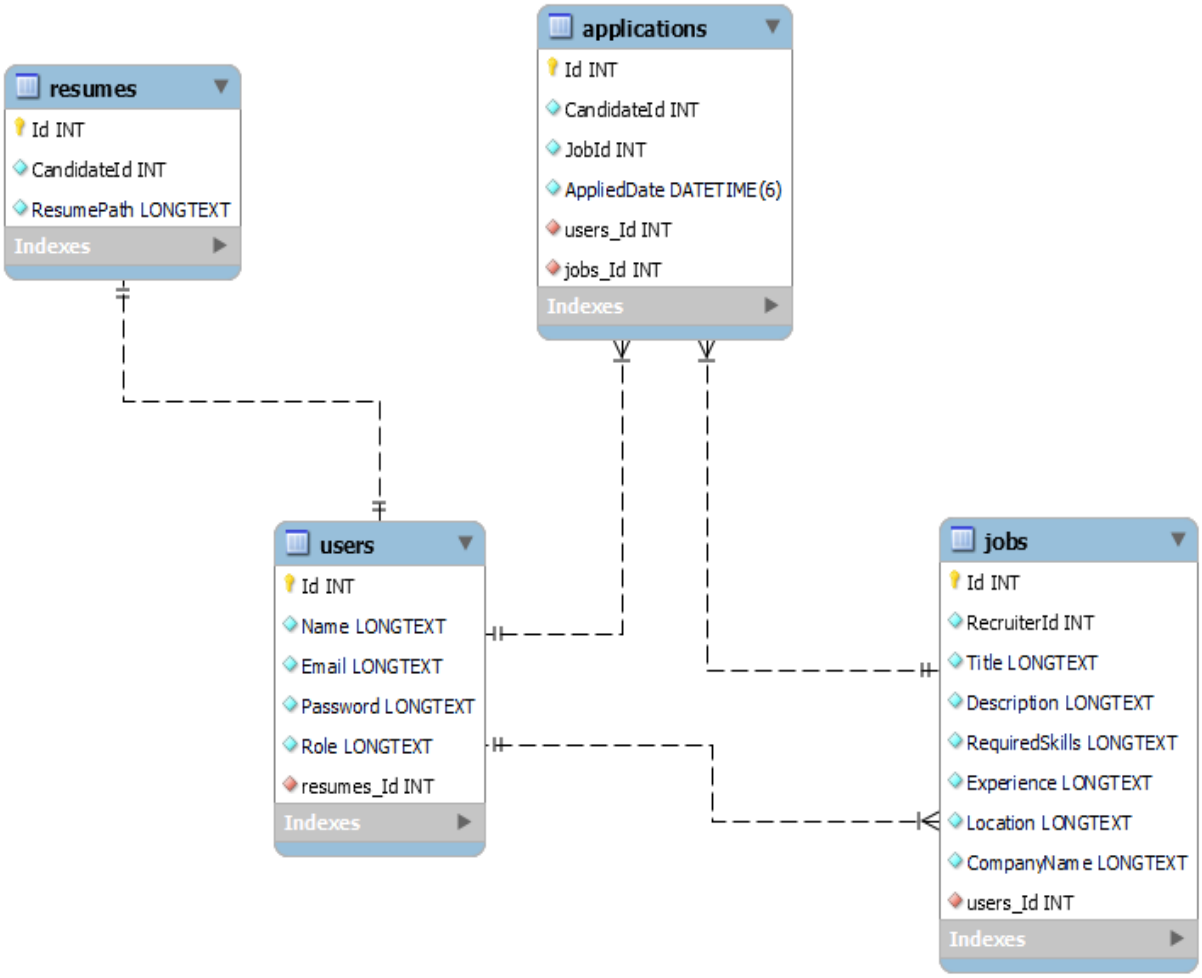


Figure 4: ER Diagram of CareerConnect Job Portal Management

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Modules

Candidate Module

The Candidate Module provides all functionalities required by job seekers. It allows users to create accounts, manage resumes, search for opportunities, and apply for suitable positions.

Candidates can easily browse available jobs, search opportunities based on location, and maintain a record of previously applied jobs. This module provides a simple and user-friendly experience for managing job applications.

Candidate Features

- Registration
- Login
- Upload Resume
- View Available Jobs
- Search Jobs by Location
- Apply for Jobs
- View Applied Jobs
- View Job Details

Screenshots

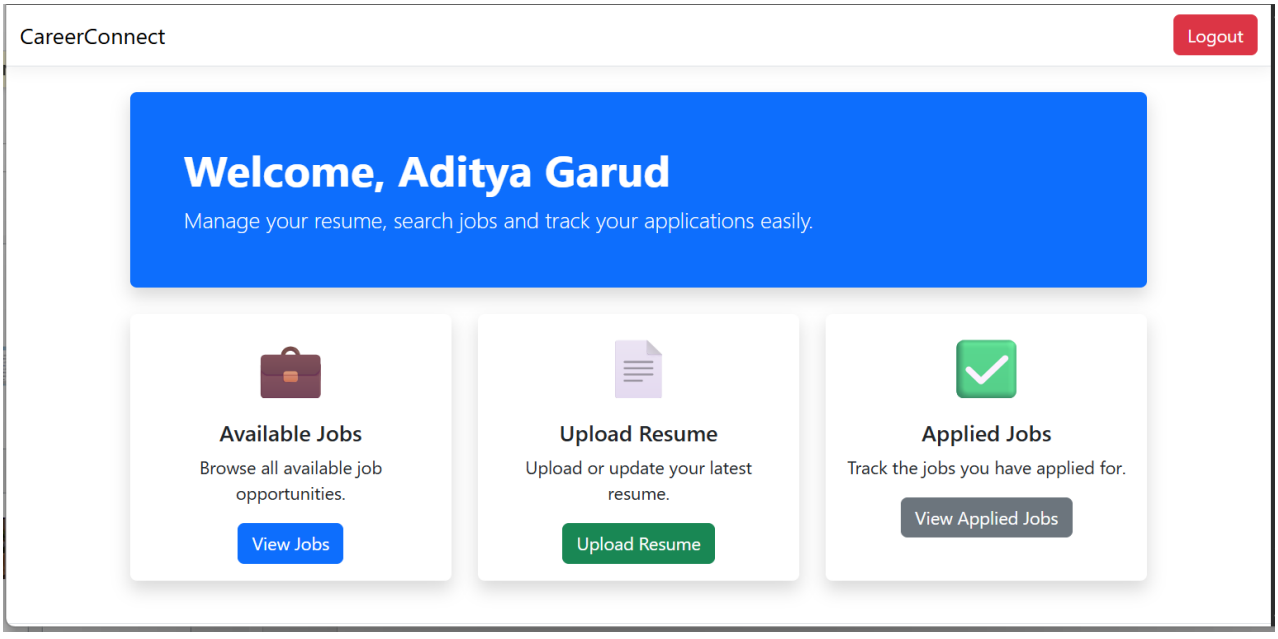


Figure 5: Candidate Dashboard

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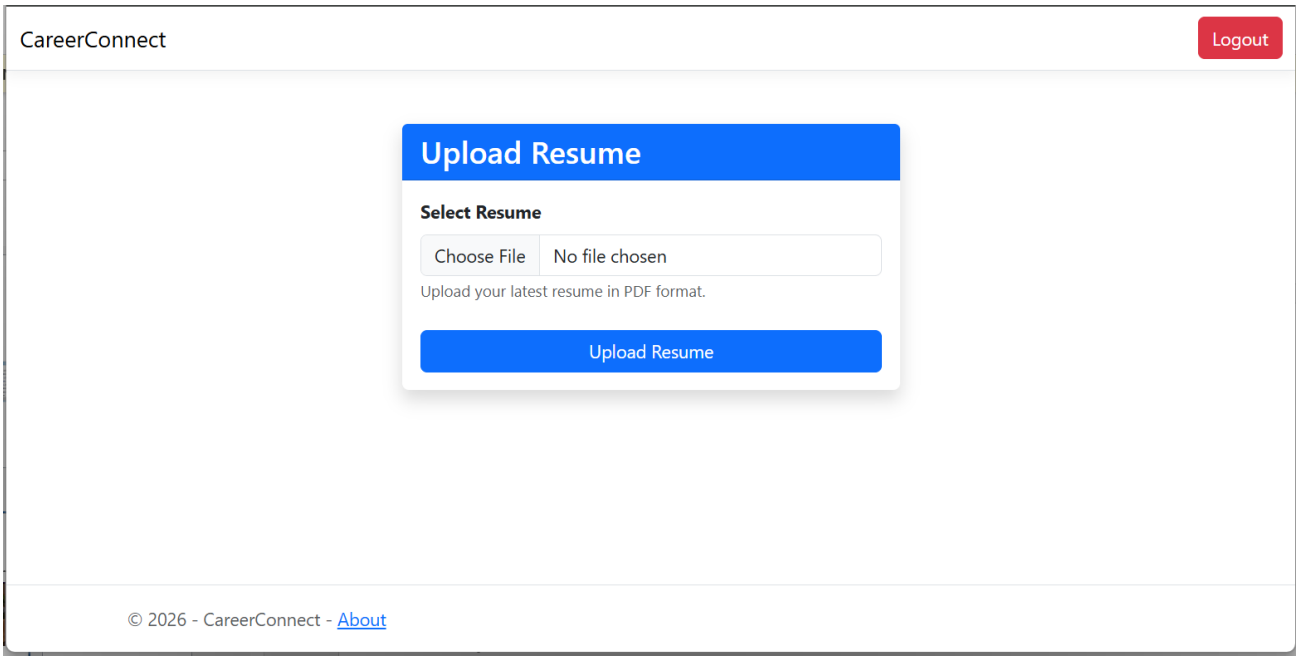


Figure 6: Upload Resume Page

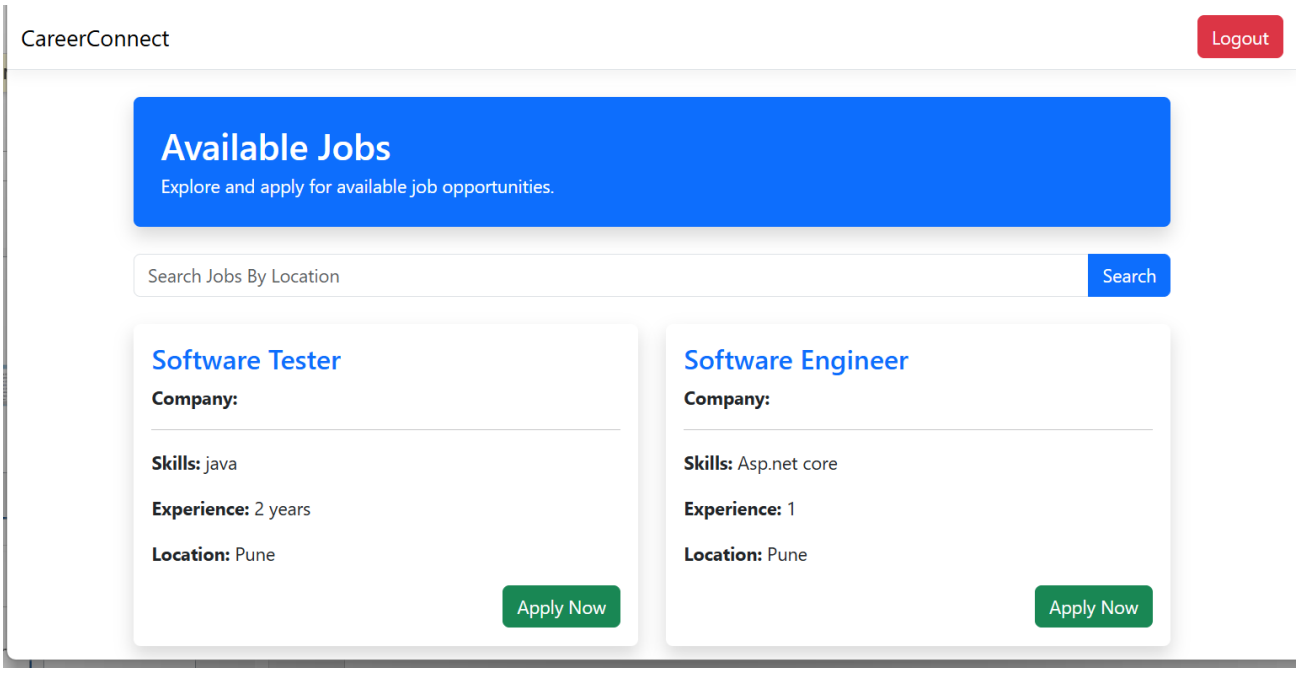


Figure 7: Available Jobs Page

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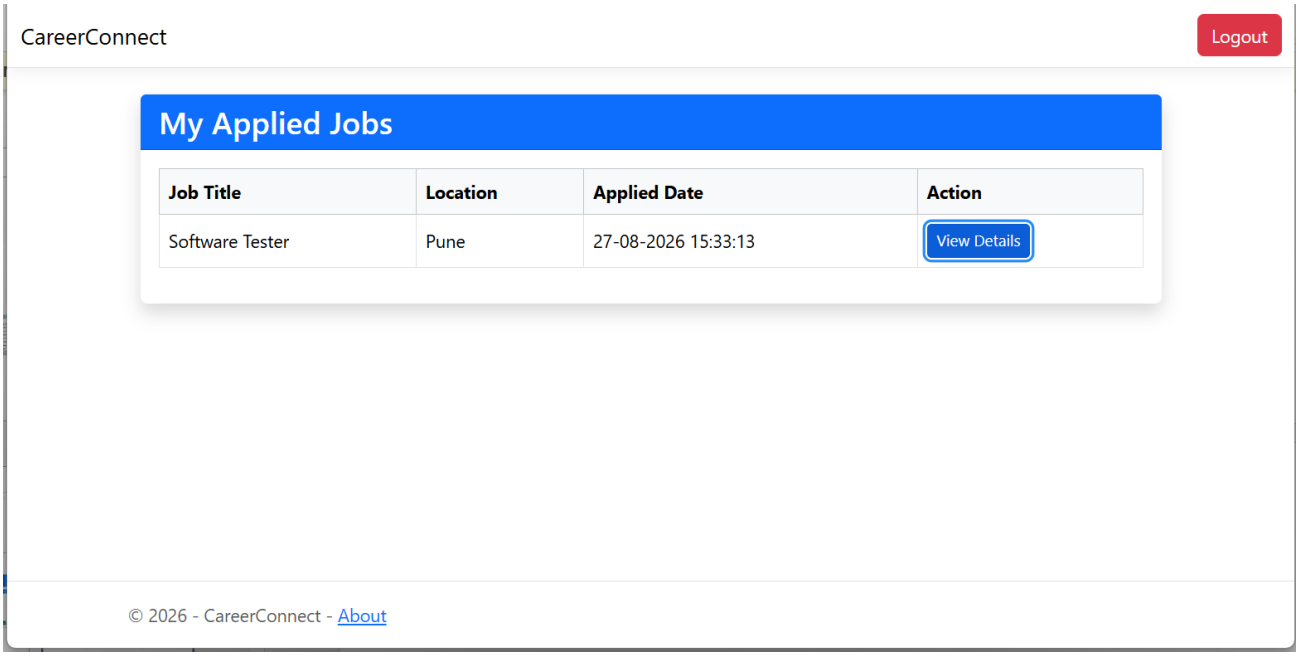


Figure 8: Applied Jobs Page

The screenshots below demonstrate the functionalities available in the Candidate Module of CareerConnect. Candidates can securely register and log in to the system, upload resumes, browse available job opportunities, search jobs based on location, apply for suitable positions, and view their previously applied jobs. These features provide a simple and user-friendly experience that helps candidates manage their job search and application activities efficiently through a centralized platform.

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Recruiter Module

The Recruiter Module helps recruiters manage recruitment activities efficiently through a centralized dashboard.

Recruiters can post new jobs, modify existing vacancies, delete obsolete job postings, review applicants, and access uploaded resumes. This module reduces manual recruitment effort and provides better visibility into the hiring process.

Recruiter Features

- Registration
- Login
- Add Job
- Edit Job
- Delete Job
- Manage Jobs
- View Applicants
- View Candidate Resumes

Screenshots

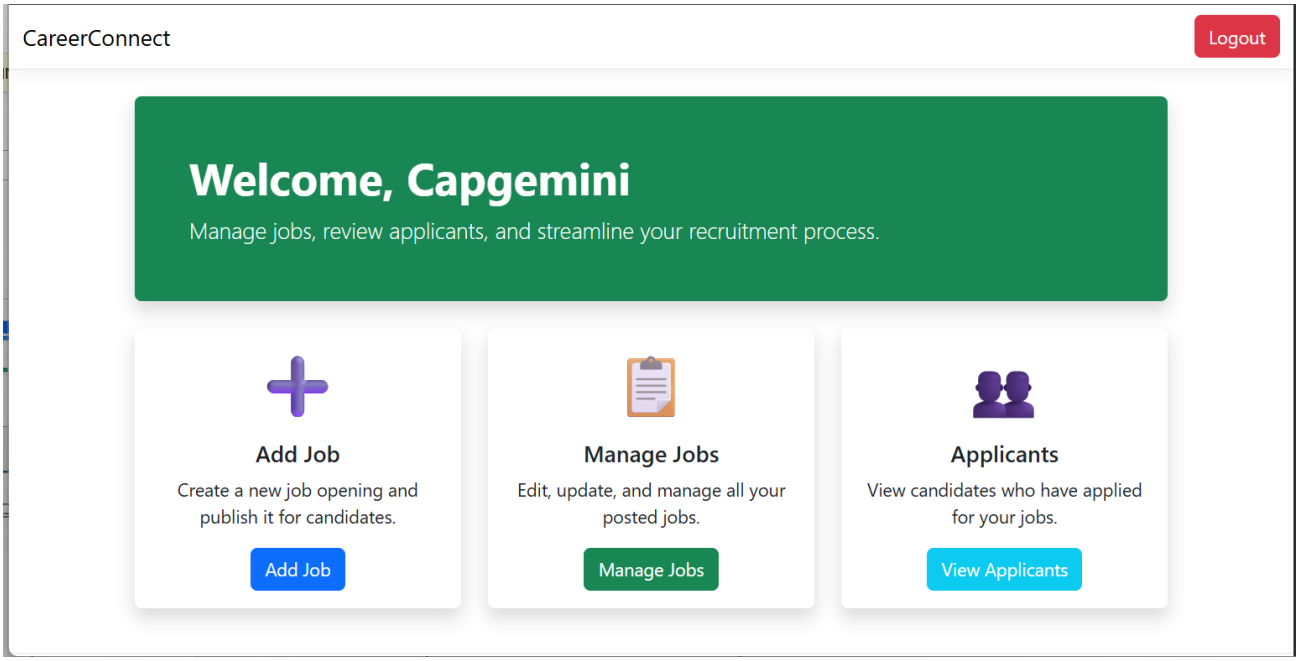


Figure 9: Recruiter Dashboard

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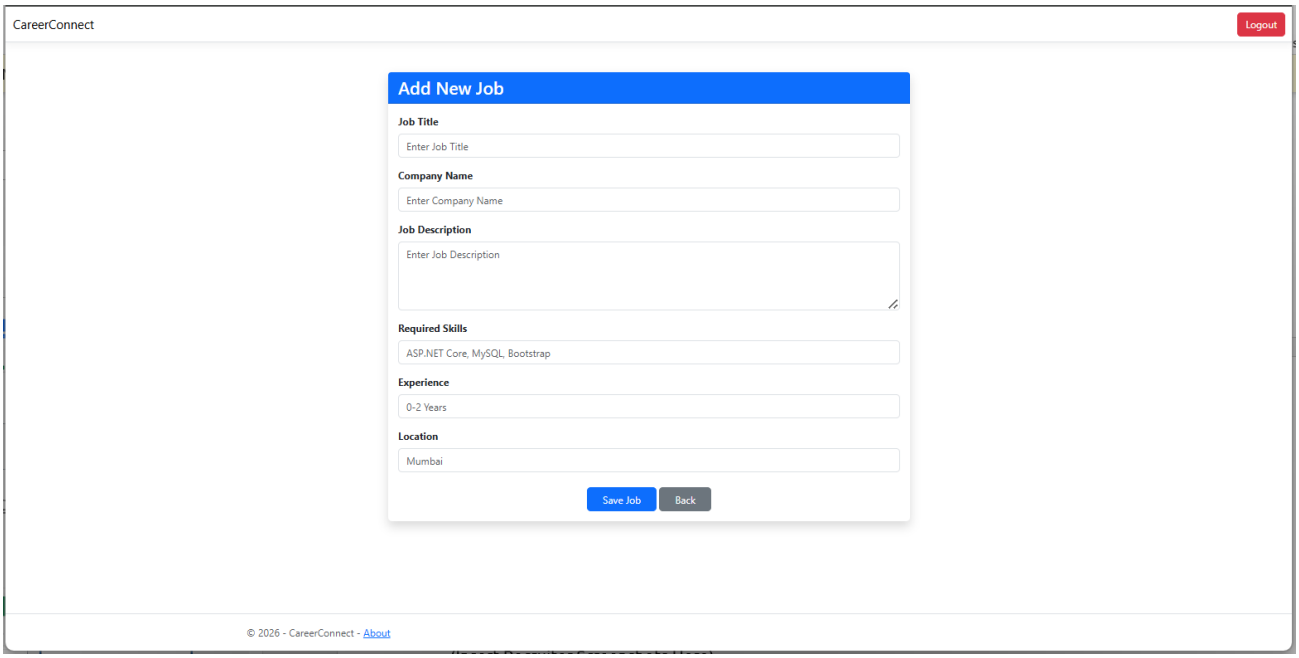


Figure 10: Add Job Page

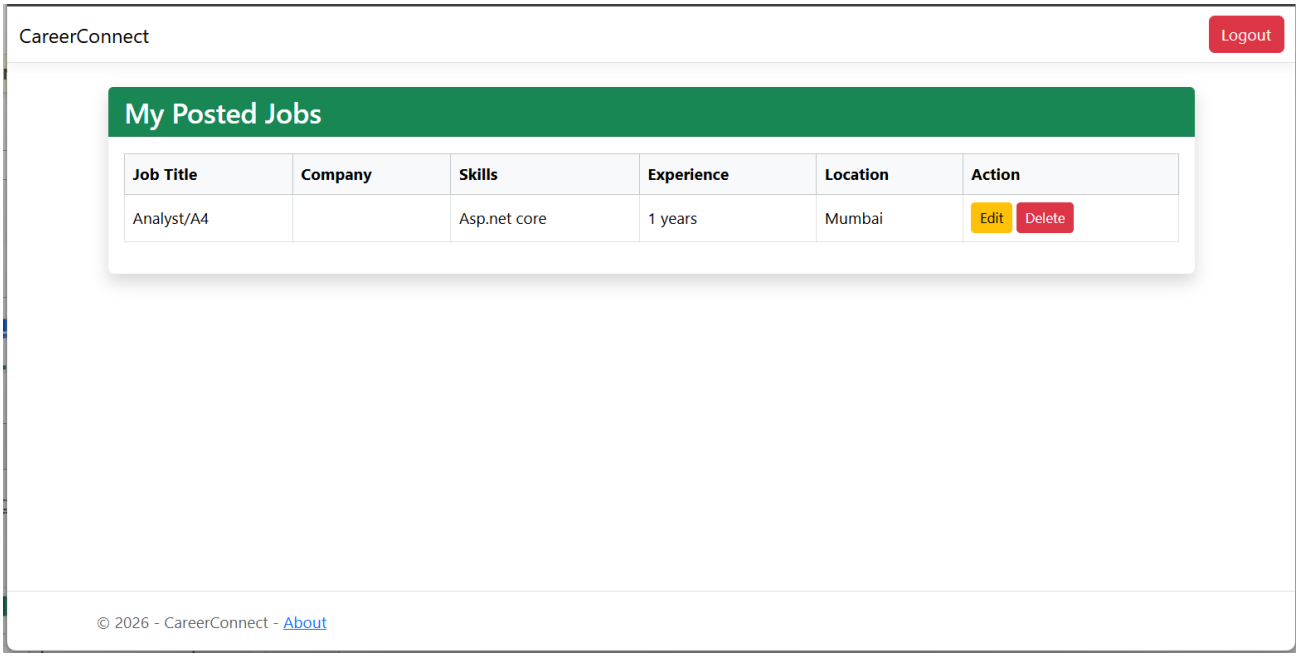


Figure 11: Manage Jobs Page

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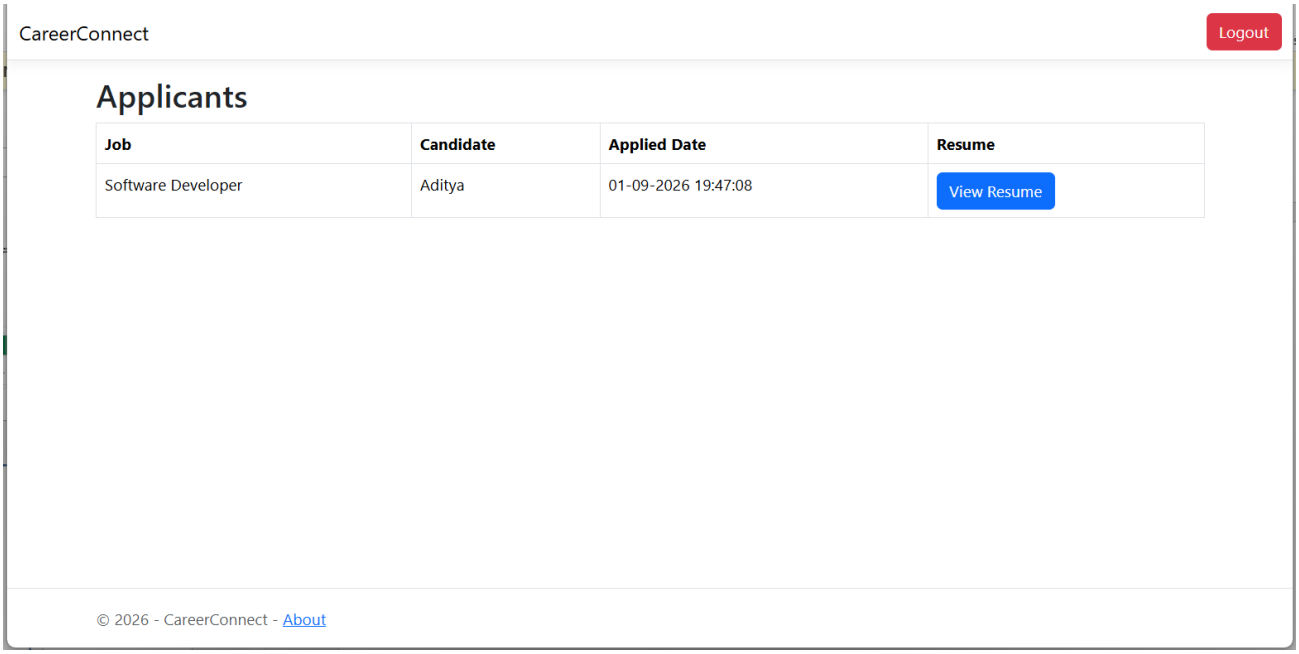


Figure 12: View Applicants Page

The screenshots below illustrate the features available in the Recruiter Module of CareerConnect. Recruiters can create and manage job postings, update vacancy information, remove outdated jobs, view applicants, and access candidate resumes through a dedicated dashboard. This module simplifies recruitment activities by providing an organized platform for managing vacancies and reviewing candidate applications efficiently.

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Development Phase

The development phase involved implementing all planned functionalities using ASP.NET Core MVC, Entity Framework Core, MySQL, and Bootstrap.

The project was developed following MVC architecture principles. Separate modules were created for Candidates and Recruiters, ensuring proper separation of responsibilities and maintainability.

Functionalities Implemented

Authentication & Authorization

- Session-Based Authentication
- Role-Based Authorization
- Login Validation

Resume Management

- Resume Upload using IFormFile
- Resume Storage in Local Directory
- Resume Path Storage in Database

Job Management

- Add Job
- Edit Job
- Delete Job
- Manage Jobs

Application Management

- View Available Jobs
- Apply for Jobs
- View Applied Jobs
- Duplicate Application Prevention

The Development Phase successfully transformed project requirements into a functional web application. It enabled the implementation of all major recruitment features and ensured smooth interaction between Candidates and Recruiters through a centralized Job Portal Management System.

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Learning Outcomes and Conclusion

Learning Outcomes

During the development of CareerConnect, practical experience was gained in multiple technologies and software development concepts.

Technologies Learned

- ASP.NET Core MVC
- C#
- Entity Framework Core
- MySQL
- Bootstrap
- HTML
- CSS

Concepts Learned

- MVC Architecture
- Session Management
- Authentication
- Authorization
- CRUD Operations
- LINQ Queries
- File Upload using IFormFile
- Database Relationships
- Software Development Life Cycle (SDLC)

Learning Outcomes

During the development of CareerConnect, practical knowledge of ASP.NET Core MVC, Entity Framework Core, MySQL, HTML, CSS, Bootstrap, and C# was gained. The project also improved understanding of important software development concepts such as MVC Architecture, Session Management, Authentication, Authorization, CRUD Operations, LINQ Queries, File Upload Handling, Database Relationships, and the Software Development Life Cycle (SDLC). Overall, the project provided valuable hands-on experience in designing, developing, and managing a complete web-based Job Portal Management System.

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Conclusion

CareerConnect is a web-based Job Portal Management System developed to provide a centralized platform for connecting job seekers and recruiters. The application simplifies recruitment activities by allowing candidates to register, upload resumes, search for available job opportunities, view job details, and apply for suitable positions through a single platform. Similarly, recruiters can create job postings, manage vacancies, view applicants, and access candidate resumes through a dedicated dashboard.

The main objective of the project was to make the job search and recruitment process more organized, efficient, and user-friendly. By bringing together candidate and recruiter functionalities within one application, CareerConnect reduces manual effort, improves accessibility to job opportunities, and helps manage recruitment-related information in a structured manner.

The project was developed using ASP.NET Core MVC, Entity Framework Core, MySQL, HTML, CSS, and Bootstrap while following the Waterfall SDLC Model. Throughout the development process, various concepts such as MVC Architecture, Session Management, Authentication, Authorization, CRUD Operations, Database Relationships, LINQ Queries, and File Upload Handling were implemented.

As a mini project developed during the BOTP training program, CareerConnect successfully achieved its intended objectives and provided valuable practical experience in web application development. The project demonstrates how modern technologies can be used to build a functional job portal that supports recruitment activities and improves the interaction between candidates and recruiters. It also serves as a foundation for future enhancements such as company profiles, interview scheduling, email notifications, and advanced job search capabilities.

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Thank You.

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